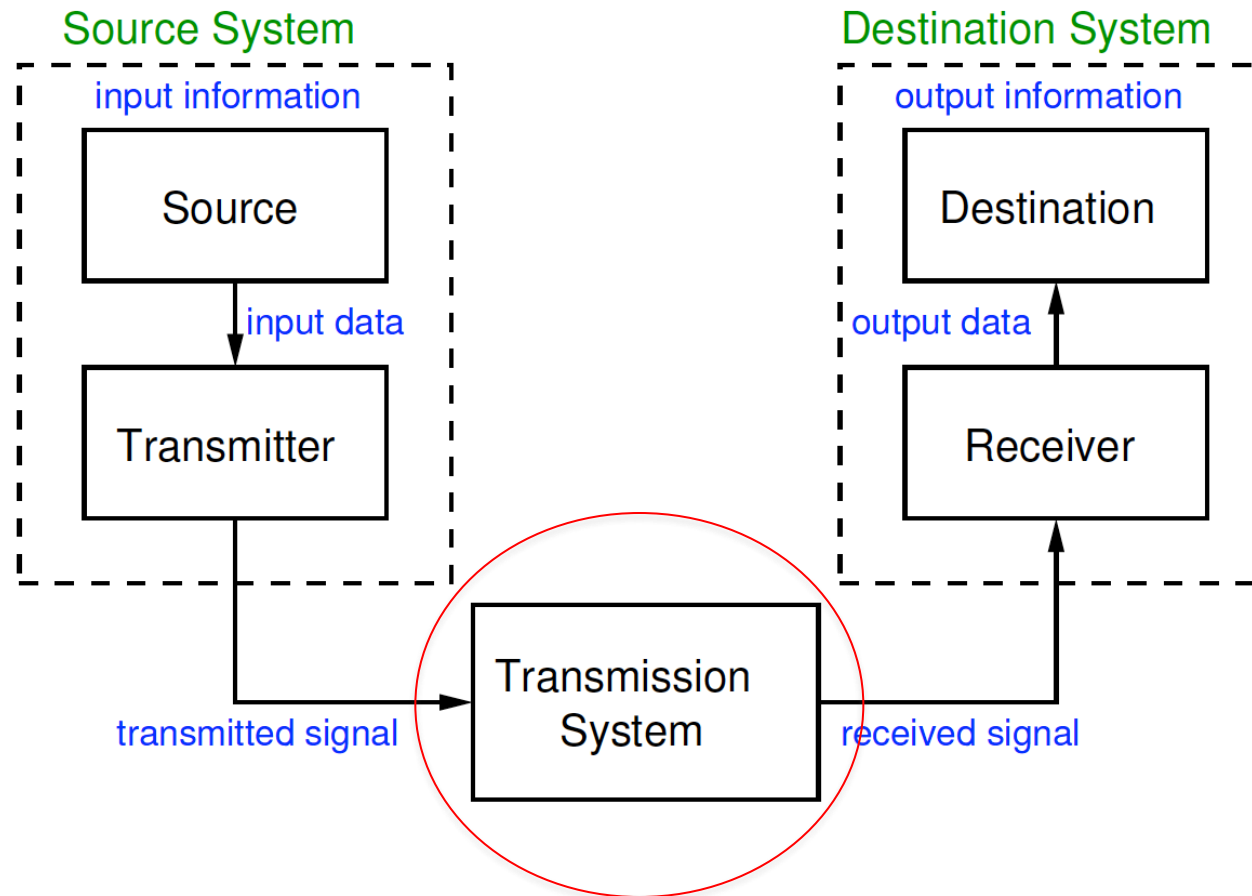


Transmission Media

Communication System

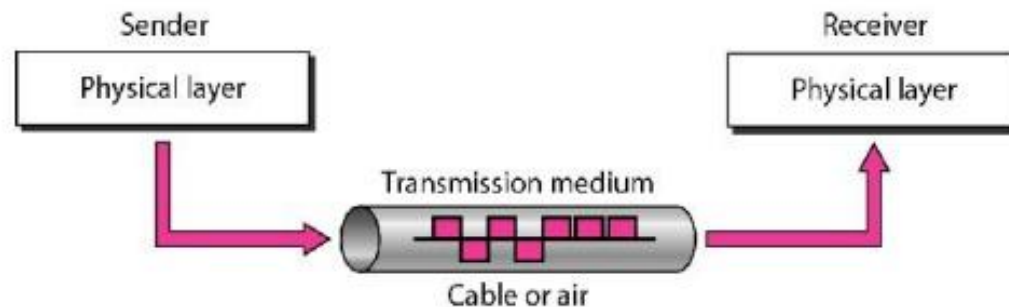


Transmission Medium

- A **transmission media** is the channel that provides the connection between the transmitter and the receiver.
 - physical or non-physical link
 - moves electromagnetic energy from one or more source to one or more receiver.
- Which medium should be used?
 - ✓ Maximize data rate
 - ✓ Maximize distance
 - ✓ Minimize transmission impairments
 - ✓ Minimize cost

Transmission Medium

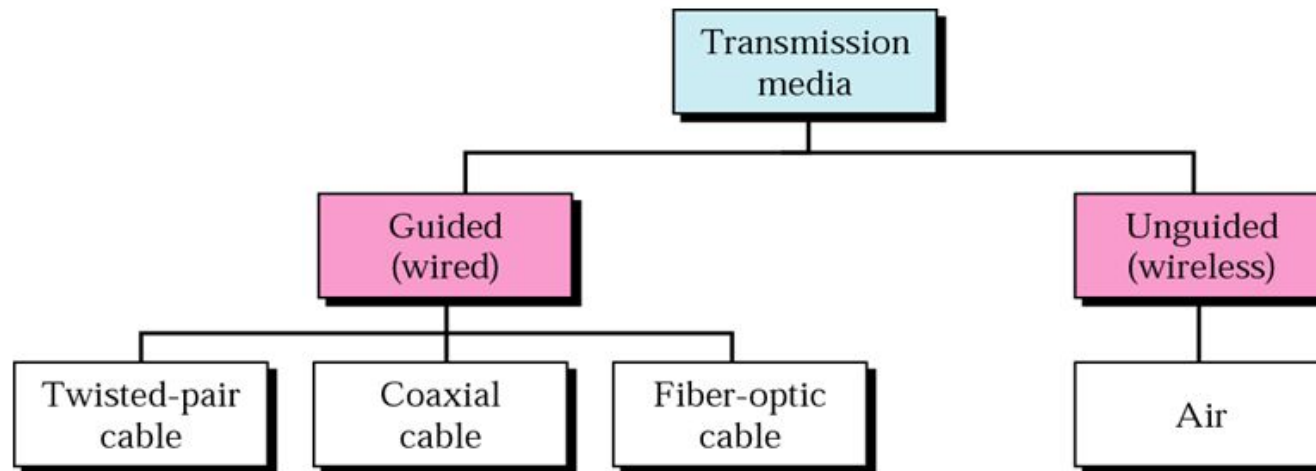
- A transmission medium can be broadly defined as anything that can carry information from a source to a destination.
- For example, the transmission medium for two people having a dinner conversation is the air. The air can also be used to convey the message in a smoke signal or semaphore.
- For a written message, the transmission medium might be a mail carrier, a truck, or an airplane.



Transmission Medium

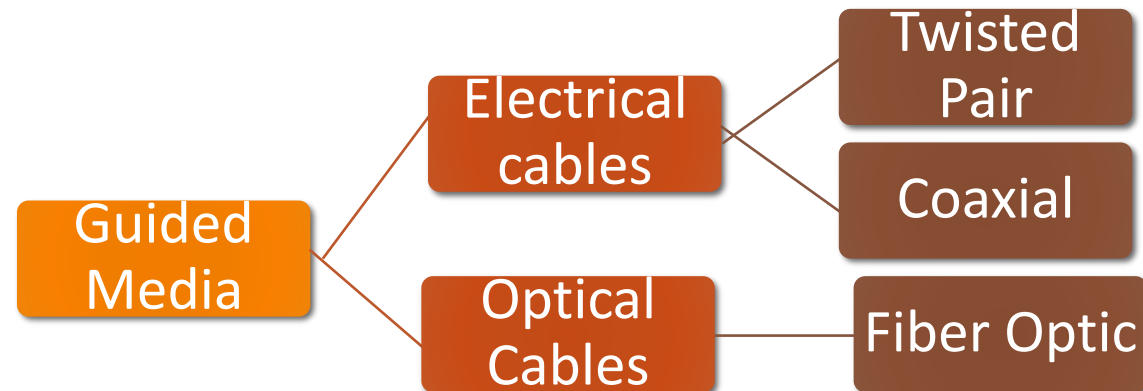
- In data communications the definition of the information and the transmission medium is more specific.
- The transmission medium is usually free space, metallic cable, or fiber-optic cable.
- The information is usually a signal that is the result of a conversion of data from another form.
- In telecommunications, transmission media can be divided into two broad categories: guided and unguided.
- **Guided (Bounded) media:** twisted-pair cable, coaxial cable, and fiber-optic cable.
- **Unguided (Unbounded) medium:** free space.

Classes of Transmission Media



Guided Media

- Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.
- A signal traveling along any of these media is directed and contained by the physical limits of the medium.
- Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current.
- Optical fiber is a cable that accepts and transports signals in the form of light.



Twisted-Pair Cable

A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in Figure



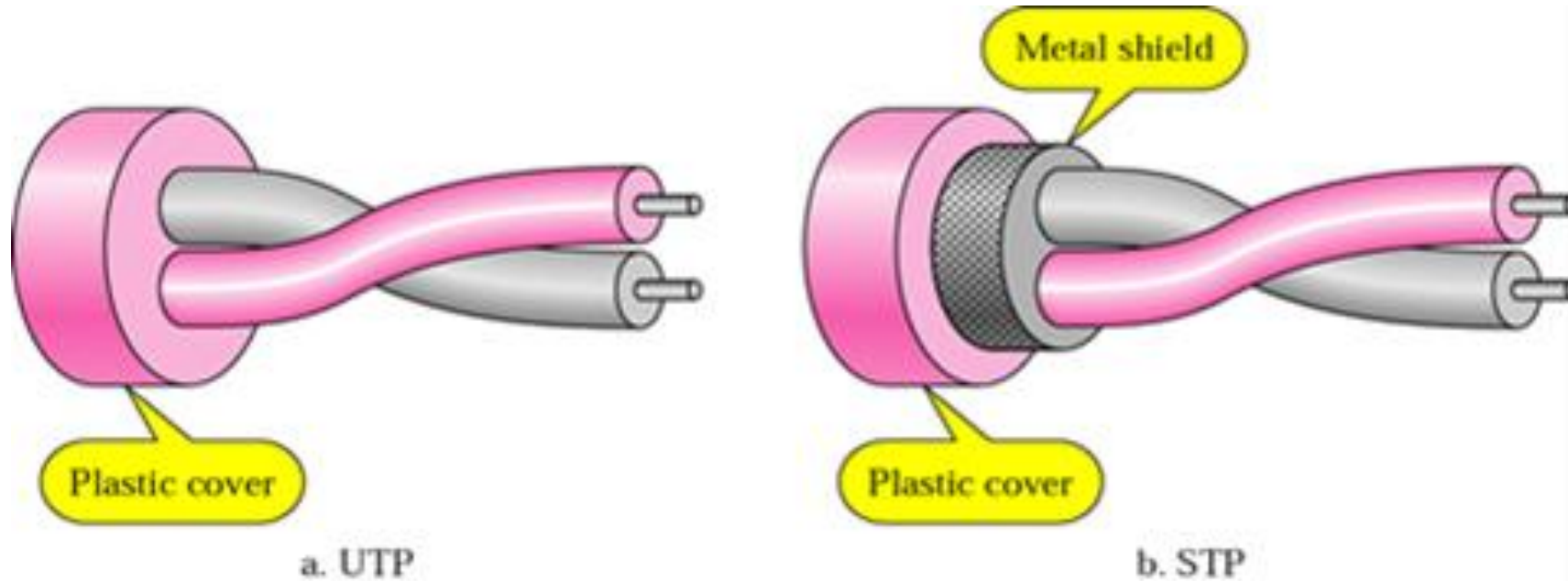
Twisted-Pair Cable

- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. The receiver uses the difference between the two.
- In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.
- If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources (e.g., one is closer and the other is farther). This results in a difference at the receiver. By twisting the pairs, a balance is maintained.
- For example, suppose in one twist, one wire is closer to the noise source and the other is farther; in the next twist, the reverse is true.
- Twisting makes it probable that both wires are equally affected by external influences (noise or crosstalk).
- This means that the receiver, which calculates the difference between the two, receives no unwanted signals. The unwanted signals are mostly canceled out.

Unshielded Versus Shielded Twisted-Pair Cable

- The most common twisted-pair cable used in communications is referred to as unshielded twisted-pair (UTP).
- IBM has also produced a version of twisted-pair cable for its use called shielded twisted-pair (STP). STP cable has a metal foil or braided mesh covering that encases each pair of insulated conductors.
- Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive. Figure shows the difference between UTP and STP.

Unshielded Versus Shielded Twisted-Pair Cable



Categories of UTP Cable

- The Electronic Industries Association (EIA) has developed standards to classify unshielded twisted-pair cable into seven categories.
- Categories are determined by cable quality, with 1 as the lowest and 7 as the highest. Each EIA category is suitable for specific uses.
- Table shows these categories.

Categories of UTP Cable

<i>Category</i>	<i>Specification</i>	<i>Data Rate (Mbps)</i>	<i>Use</i>
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T-lines	2	T-llines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs
SE	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called SSTP (shielded screen twisted-pair). Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk: and increases the data rate.	600	LANs

Connector

The most common UTP connector is RJ45 (RJ stands for registered jack), as shown in Figure. The RJ45 is a keyed connector, meaning the connector can be inserted in only one way.



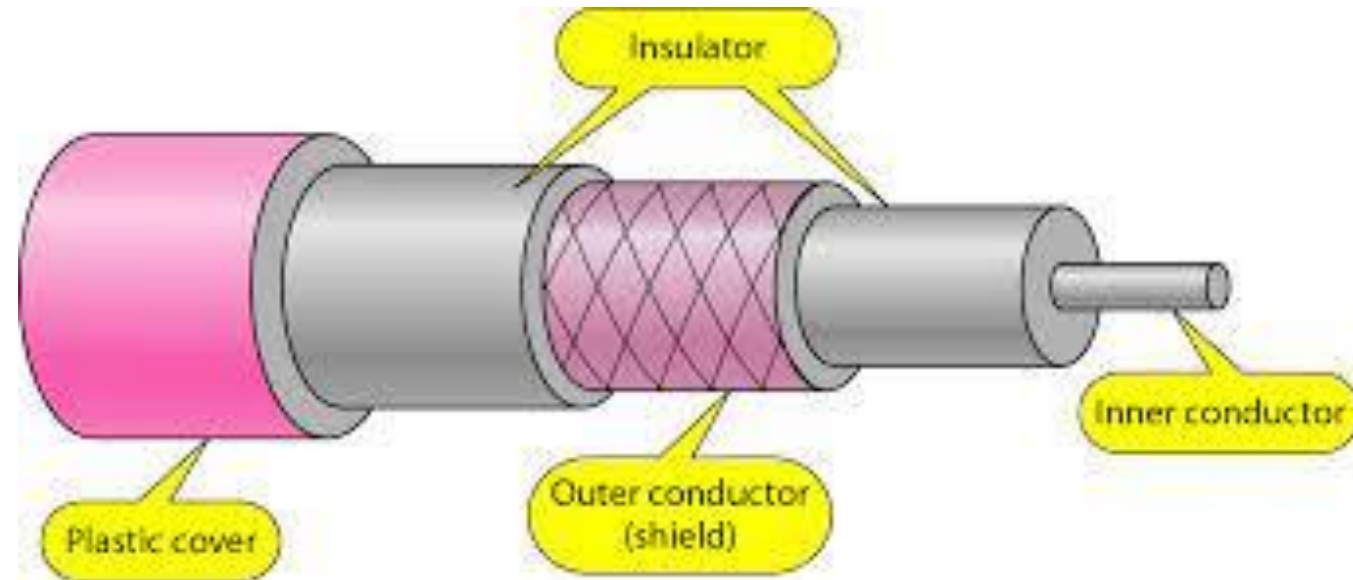
Applications of UTP

- Twisted-pair cables are used in telephone lines to provide voice and data channels.
- The local loop-the line that connects subscribers to the central telephone office---commonly consists of unshielded twisted-pair cables.
- The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.
- Local-area networks, such as 10Base-T and 100Base-T, also use twisted-pair cables.

Coaxial Cable

- ❑ Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable, in part because the two media are constructed quite differently.
- ❑ Instead of having two wires, coax has a central core conductor of solid or stranded wire (usually copper) enclosed in an insulating sheath, which is, in turn, encased in an outer conductor of metal foil, braid, or a combination of the two.
- ❑ The outer metallic wrapping serves both as a shield against noise and as the second conductor, which completes the circuit.
- ❑ This outer conductor is also enclosed in an insulating sheath, and the whole cable is protected by a plastic cover

Coaxial Cable



Coaxial Cable Standards

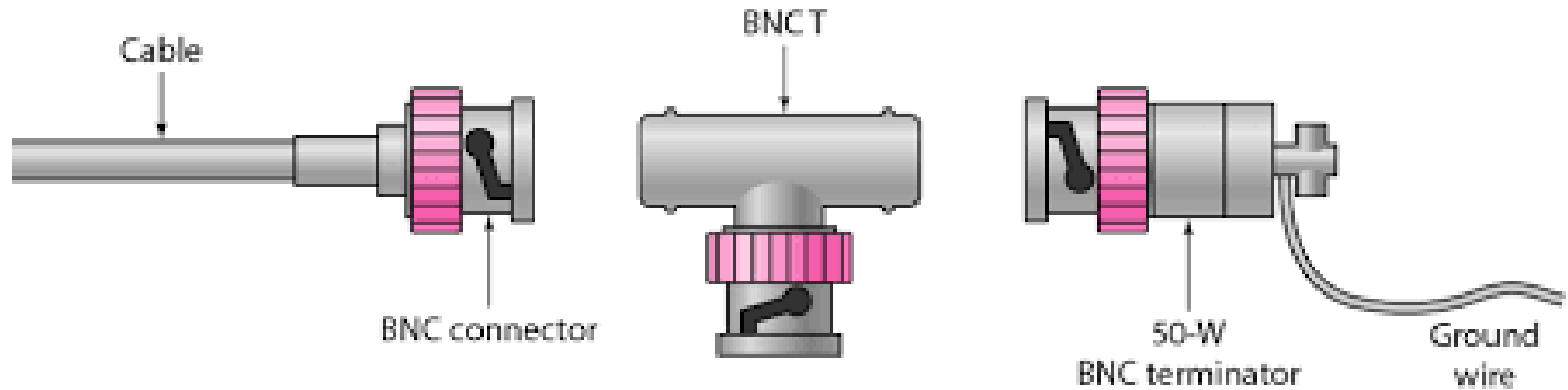
- ❑ Coaxial cables are categorized by their radio government (RG) ratings.
- ❑ Each RG number denotes a unique set of physical specifications, including the wire gauge of the inner conductor, the thickness and type of the inner insulator, the construction of the shield, and the size and type of the outer casing.
- ❑ Each cable defined by an RG rating is adapted for a specialized function, as shown in Table

<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Coaxial Cable Connectors

- ❑ To connect coaxial cable to devices, we need coaxial connectors. The most common type of connector used today is the Bayone-Neill-Concelman (BNC), connector.
- ❑ Figure shows three popular types of these connectors: the BNC connector, the BNC T connector, and the BNC terminator.
- ❑ The BNC connector is used to connect the end of the cable to a device, such as a TV set. The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.
- ❑ The BNC terminator is used at the end of the cable to prevent the reflection of the signal.

Coaxial Cable Connectors



Coaxial Cable

Provide much more shielding from interference than twisted pair:

- Higher data rates;

- more devices on a shared line;

- Longer distances.

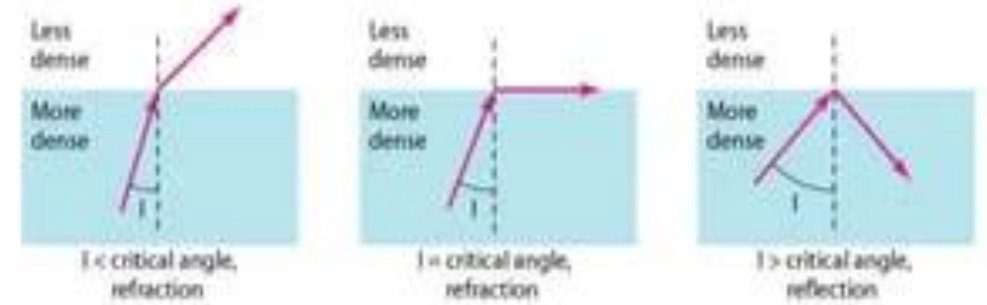
Widely used for cable TV, as well as other audio/video cabling.

Used in long-distance telecommunications, although optical fiber is more relevant now

Fiber Optic Cable

- ❑ A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- ❑ To understand optical fiber, we first need to explore several aspects of **the nature of light**.
- ❑ **Light travels in a straight line as long as it is moving through a single uniform substance.**
- ❑ If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction.
- ❑ Figure shows how a ray of light changes direction when going from a more dense to a less dense substance.

Fiber Optic Cable



As the figure shows,

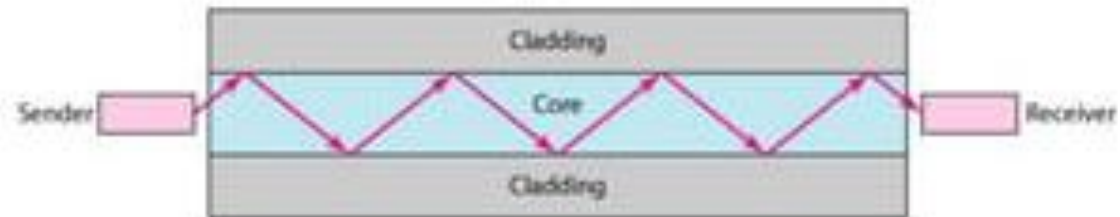
A. if the angle of incidence i (the angle the ray makes with the line perpendicular to the interface between the two substances) is less than the critical angle, the ray refracts and moves closer to the surface.

B. If the angle of incidence is equal to the critical angle, the light bends along the interface.

C, If the angle is greater than the critical angle, the ray reflects (makes a turn) and travels again in the denser substance.

Fiber Optic Cable

- ❑ Optical fibers use **reflection to guide light through a channel.**
- ❑ A glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- ❑ The **difference in density of the two materials** must be such that a beam of **light moving through the core is reflected off the cladding instead of being refracted into it.**



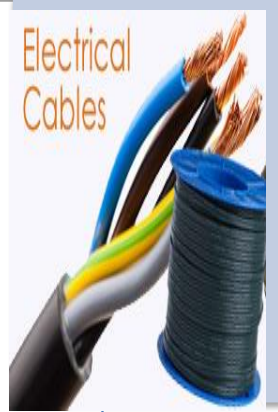
Fiber Optic Cable: Advantages

- ❑ **Higher bandwidth.** Fiber-optic cable can support dramatically higher bandwidths (and hence data rates) than either twisted-pair or coaxial cable. Currently, data rates and bandwidth utilization over fiber-optic cable are limited not by the medium but by the signal generation and reception technology available.
- ❑ **Less signal attenuation.** Fiber-optic transmission distance is significantly greater than that of other guided media. A signal can run for 50 km without requiring regeneration. We need repeaters every 5 km for coaxial or twisted-pair cable.
- ❑ **Immunity to electromagnetic interference.** Electromagnetic noise cannot affect fiber-optic cables.
- ❑ **Resistance to corrosive materials.** Glass is more resistant to corrosive materials than copper.
- ❑ **Light weight.** Fiber-optic cables are much lighter than copper cables.
- ❑ **Greater immunity to tapping.** Fiber-optic cables are more immune to tapping than copper cables. Copper cables create antenna effects that can easily be tapped.

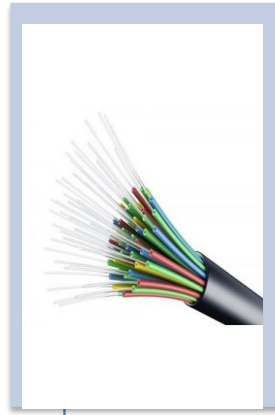
Fiber Optic Cable: Disadvantages

- ❑ **Installation and maintenance.** Fiber-optic cable is a relatively new technology. Its installation and maintenance require expertise that is not yet available everywhere.
- ❑ **Unidirectional light propagation.** Propagation of light is unidirectional. If we need bidirectional communication, two fibers are needed.
- ❑ **Cost.** The cable and the interfaces are relatively more expensive than those of other guided media. If the demand for bandwidth is not high, often the use of optical fiber cannot be justified.

Comparison of Guided Media



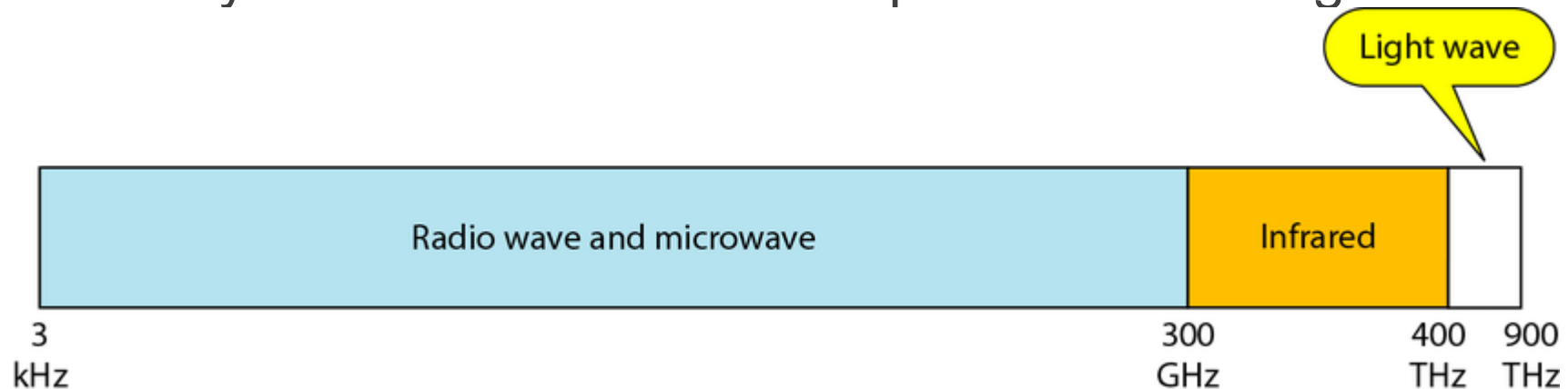
- Moderate data rates: 1Gb/s
- Maximum distance: 2km (twisted pair); 10km (coaxial)
- Cheapest for low data rates
- UTP: easy to install, susceptible to interference
- STP, Coaxial Cable: rigid, protection against interference



- Very high data rates: 100Gb/s+
- Maximum distance: 40km
- Expensive equipment, but cost effective for high data rates
- Difficult to install

Unguided Media

- ❑ Unguided media transport electromagnetic waves without using a physical conductor.
- ❑ This type of communication is often referred to as wireless communication.
- ❑ Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.



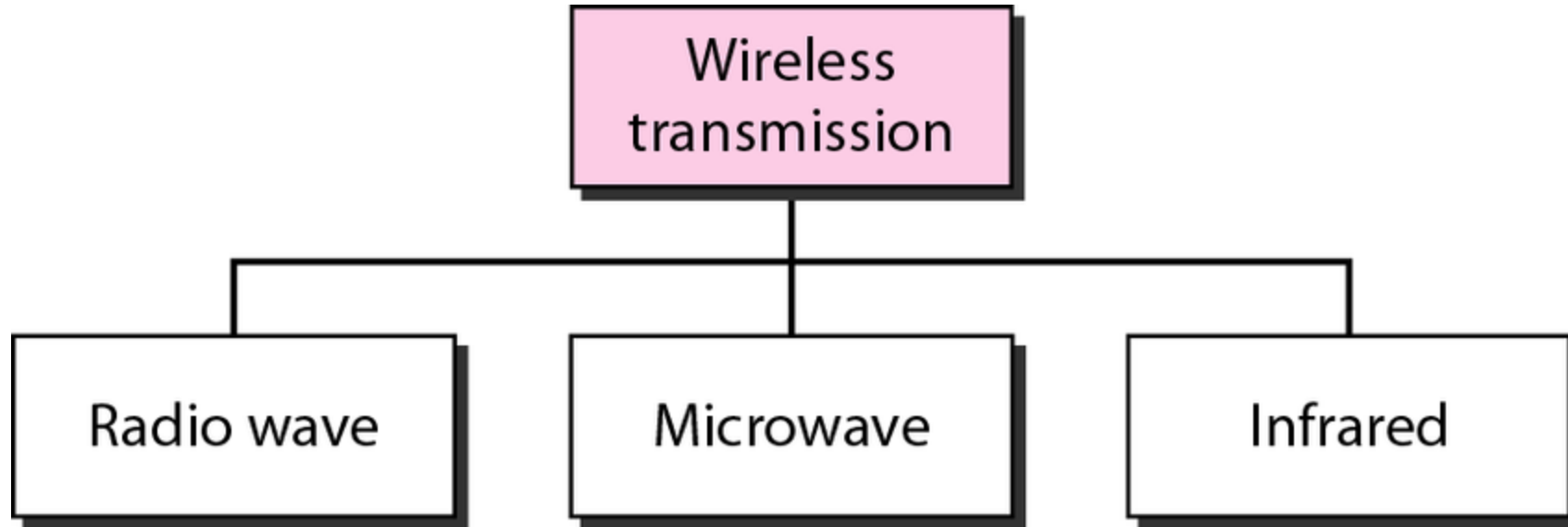
Unguided Media

- ❑ Unguided signals can travel from the source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation.
- ❑ In ground propagation, radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- ❑ These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance.
- ❑ In sky propagation, higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth.
- ❑ This type of transmission allows for greater distances with lower output power. **In line-of-sight propagation, very high-frequency signals are transmitted in straight lines directly from antenna to antenna.**
- ❑ **Antennas** must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth.
- ❑ **Line-of-sight propagation** is tricky because radio transmissions cannot be completely focused.

Unguided Media

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3-30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30-300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz-3 MHz	Sky	AM radio
HF (high frequency)	3-30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30-300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz-3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3-30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30-300 GHz	Line-of-sight	Radar, satellite

Wireless Transmission Waves



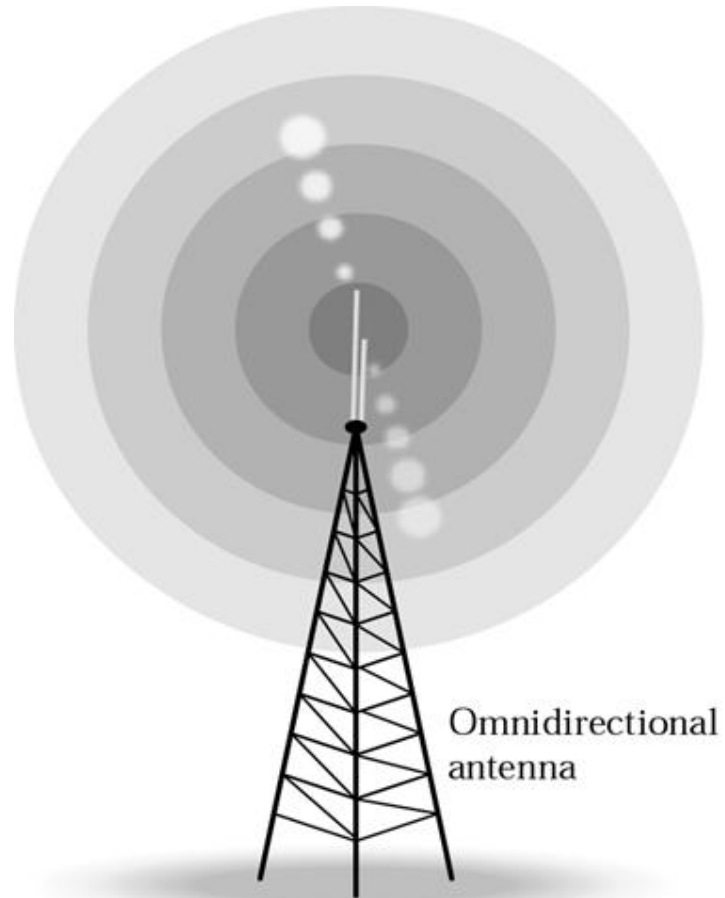
Radio Waves

- ❑ Although there is no clear-cut demarcation between radio waves and microwaves, electromagnetic waves ranging in frequencies between **3 kHz and 1 GHz are normally called radio waves; waves ranging in frequencies between 1 and 300 GHz are called microwaves.**
- ❑ However, the behavior of the waves, rather than the frequencies, is a better criterion for classification.
- ❑ Radio waves are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned.
- ❑ A sending antenna sends waves that can be received by any receiving antenna. The omnidirectional property has a disadvantage, too.

Radio Waves

- ❑ The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.
- ❑ **Radio waves, particularly those waves that propagate in the sky mode, can travel long distances.** This makes radio waves a good candidate for long-distance broadcasting such as AM radio. Radio waves, particularly those of low and medium frequencies, can penetrate walls.
- ❑ The radio wave band is relatively narrow, just under 1 GHz, compared to the microwave band. When this band is divided into subbands, the subbands are also narrow, leading to a low data rate for digital communications.
- ❑ **Applications:** AM and FM radio, television, cordless phones, and paging are examples of multicasting.

Radio Waves



Microwave

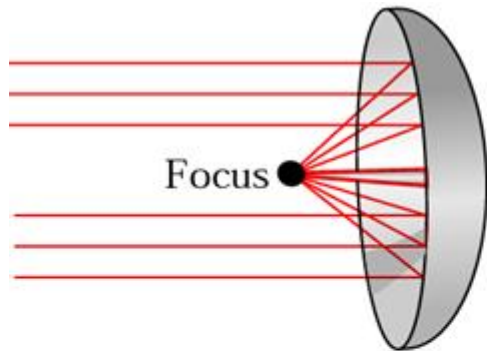
- Electromagnetic waves having frequencies between **1 and 300 GHz** are called **microwaves**.
- **Microwaves are unidirectional.** When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.
- The unidirectional property has an obvious advantage.
- A pair of antennas can be aligned without interfering with another pair of aligned antennas. The following describes some characteristics of microwave propagation:

Microwave

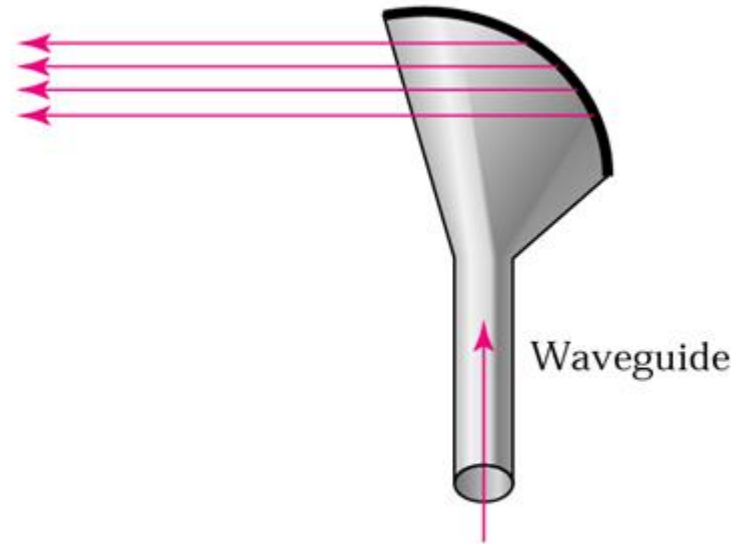
- **Microwave propagation is line-of-sight:** Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall. The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves.
- Repeaters are often needed for longdistance communication.
- **Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.**
- The microwave band is relatively wide, almost 299 GHz. Therefore wider subbands can be assigned, and a high data rate is possible.
- Use of certain portions of the band requires permission from authorities.

Unidirectional Antenna

Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish and the horn (see Figure).



a. Dish antenna



b. Horn antenna

Unidirectional Antenna

- **A parabolic dish antenna is based on the geometry of a parabola:** Every line parallel to the line of symmetry (line of sight) reflects off the curve at angles such that all the lines intersect in a common point called the focus.
- The parabolic dish works as a funnel, catching a wide range of waves and directing them to a common point. In this way, more of the signal is recovered than would be possible with a single-point receiver.
- Outgoing transmissions are broadcast through a horn aimed at the dish. The microwaves hit the dish and are deflected outward in a reversal of the receipt path.
- **A horn antenna looks like a gigantic scoop. Outgoing transmissions are broadcast up a stem (resembling a handle) and deflected outward in a series of narrow parallel beams by the curved head.**
- Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem.

Applications Of Microwave

- Microwaves, due to their unidirectional properties, are very useful when unicast (one-to-one) communication is needed between the sender and the receiver.
- They are used in cellular phones, satellite networks, and wireless LANs.

Infrared

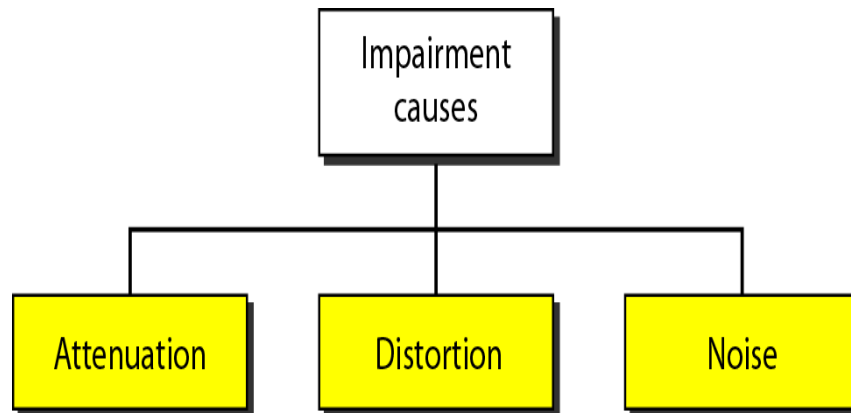
- Infrared waves, with frequencies from **300 GHz to 400 THz (wavelengths from 1 mm to 770 nm)**.
- It can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls.
- This advantageous characteristic prevents interference between one system and another; **a short-range communication system in one room cannot be affected by another system in the next room.**
- When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication.
- In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Applications of Infrared

- The infrared band, almost 400 THz, has an excellent potential for data transmission. Such a wide bandwidth can be used to transmit digital data with a very high data rate.
- The Infrared Data Association (IrDA), an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.
- For example, some manufacturers provide a special port called the IrDA port that allows a wireless keyboard to communicate with a PC.
- The standard originally defined a data rate of 75 kbps for a distance up to 8 m. The recent standard defines a data rate of 4 Mbps.
- Infrared signals defined by IrDA transmit through line of sight; the IrDA port on the keyboard needs to point to the PC for transmission to occur.

TRANSMISSION IMPAIRMENT

- Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment.
- **This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.**
- What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise.

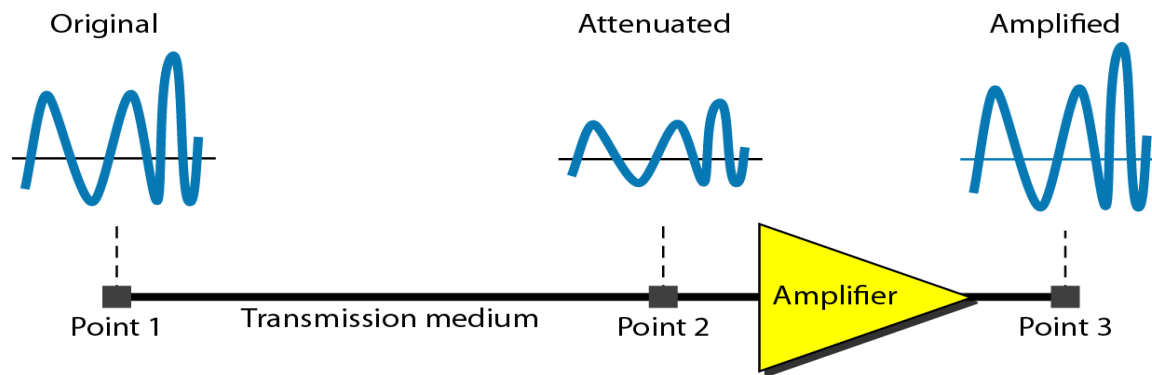


1. Attenuation

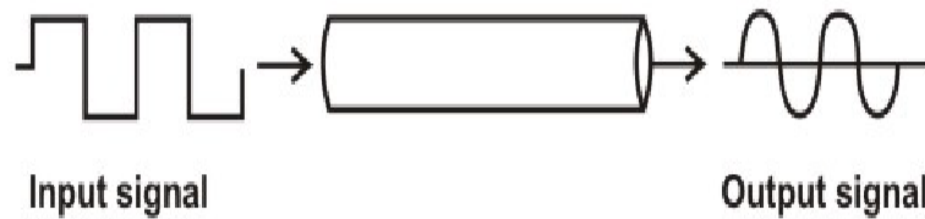
- Attenuation means a **loss of energy**.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- **That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat.**
- To compensate for this loss, amplifiers are used to amplify the signal. Figure shows the effect of attenuation and amplification.

1. Attenuation

1- Attenuated signal cannot be detected and interpreted properly → Amplifier



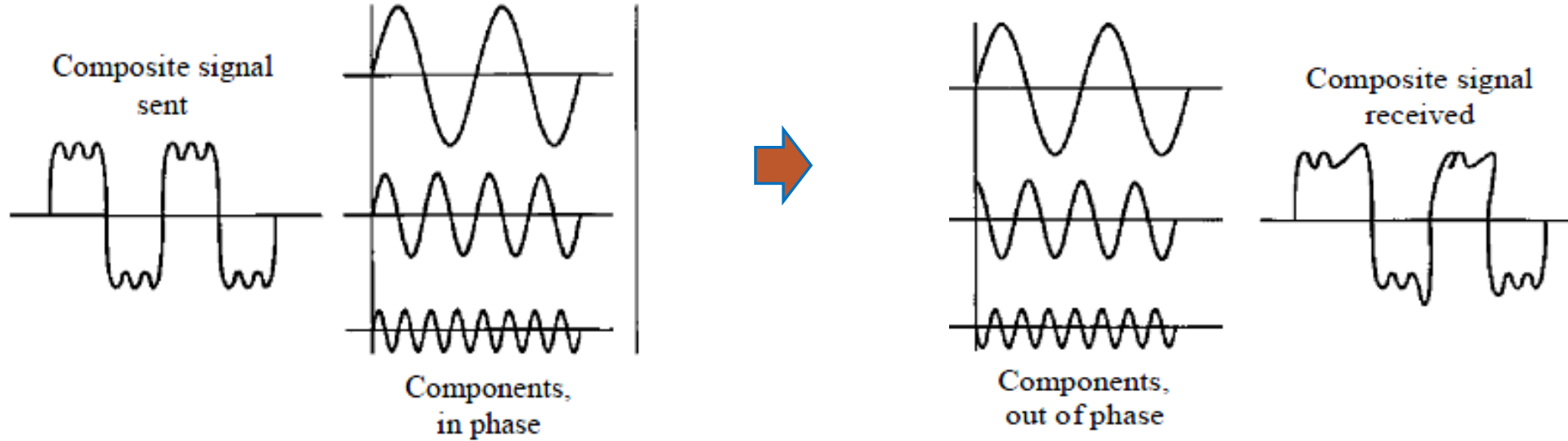
2. Attenuation Distortion



2. Distortion

- Distortion means that the **signal changes its form or shape**.
- Distortion can occur in a composite signal made of different frequencies.
- **Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination.**
- Differences in delay may create a difference in phase if the delay is not exactly the same as the period duration.
- In other words, signal components at the receiver have phases different from what they had at the sender.
- The shape of the composite signal is therefore not the same. Figure shows the effect of distortion on a composite signal.

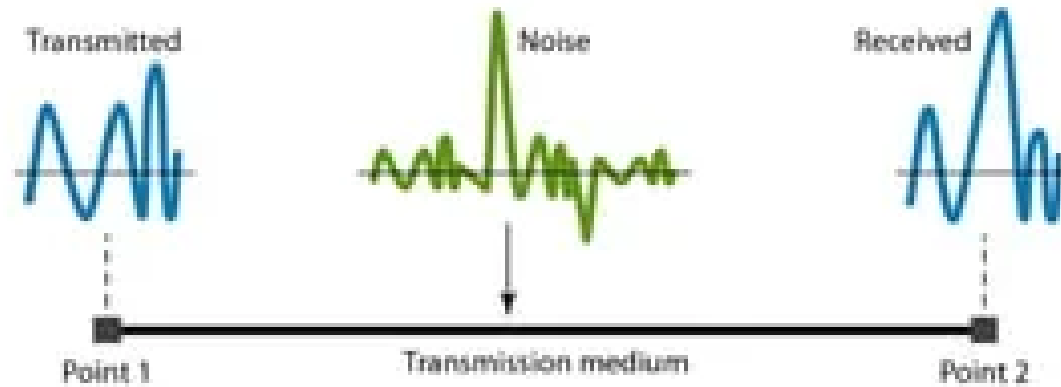
2. Distortion



3. Noise

- **Noise** is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- **Thermal noise** is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
- **Induced noise** comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna.
- **Impulse noise** is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.
- Figure shows the effect of noise on a signal.

3. Noise



Signal-to-Noise Ratio (SNR)

In the study of noise, it is not important to know the absolute value of noise.

Even if the power of the noise is very small, it may have a significant effect if the power of the signal is also small.

What is important is a comparison between noise and the signal.

The *signal-to-noise ratio* (SNR) is the ratio of signal power to noise power.

$$\text{SNR} = P_s / P_n$$

Example:

$$P_{s1} = 5 \text{ w}$$

$$P_{s2} = 50,000 \text{ w}$$

$$P_{n1} = 50 \text{ mw}$$

$$P_{n2} = 50 \text{ w}$$

$$\begin{aligned} \text{SNR}_1 &= 5 / 50 \times 10^{-3} \\ &= 1,000 \end{aligned}$$

$$\begin{aligned} \text{SNR}_2 &= 50,000 / 50 \\ &= 1,000 \end{aligned}$$

Both systems have the same performance.

Signal-to-Noise Ratio (SNR)

Ideally, $SNR = \infty$ (when $P_n = 0$).

In practice, SNR should be high as possible.

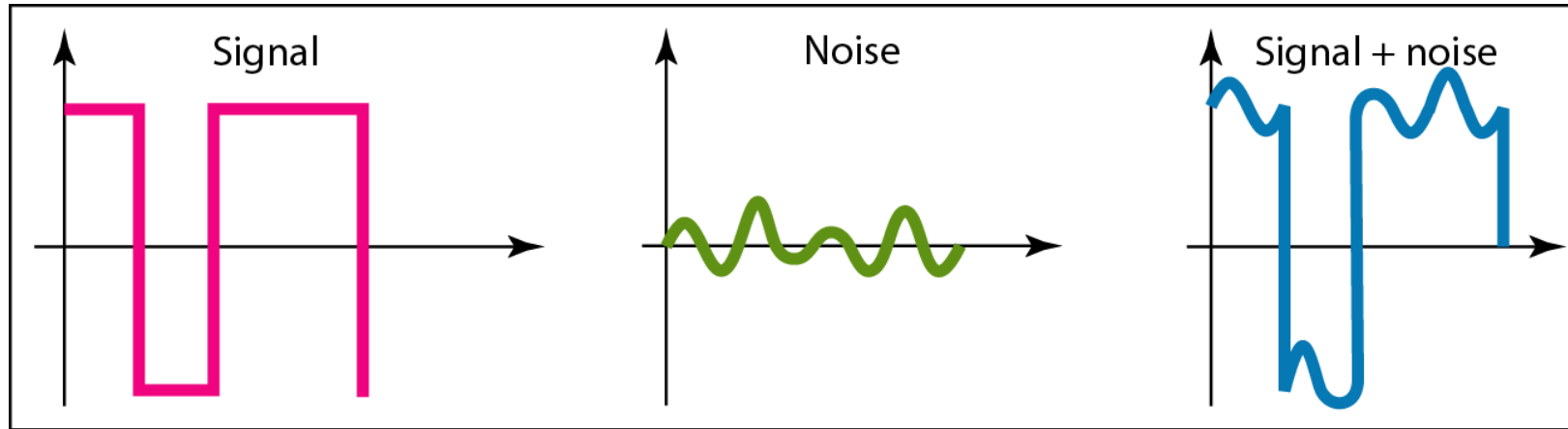
A high SNR ratio means a good-quality signal.

A low SNR ratio means a low-quality signal.

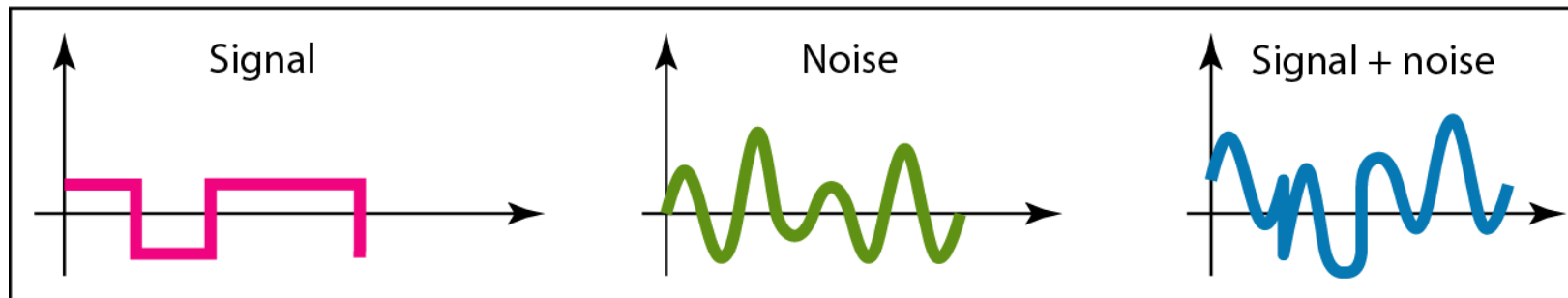
The SNR is normally expressed in decibels, that is:

$$SNR = 10 \log_{10} (P_s / P_n) \quad dB$$

Figure: Two cases of SNR: a high SNR and a low SNR



a. Large SNR



b. Small SNR

Example:

- The power of a signal is 10 mW and the power of the noise is 1 μ W; what are the values of SNR and SNR_{dB}

$$\text{SNR} = 10 \times 10^{-3} / 10^{-6} = 10,000$$

$$\begin{aligned}\text{SNR}_{\text{dB}} &= 10 \log_{10} (10 \times 10^{-3} / 10^{-6}) \\ &= 10 \log_{10} (10,000) = 40 \text{ dB}\end{aligned}$$

Example

- If the SNR_{dB} is 36 , find the value of SNR?

$$\text{SNR}_{\text{dB}} = 10 \times \log_{10}(\text{SNR})$$

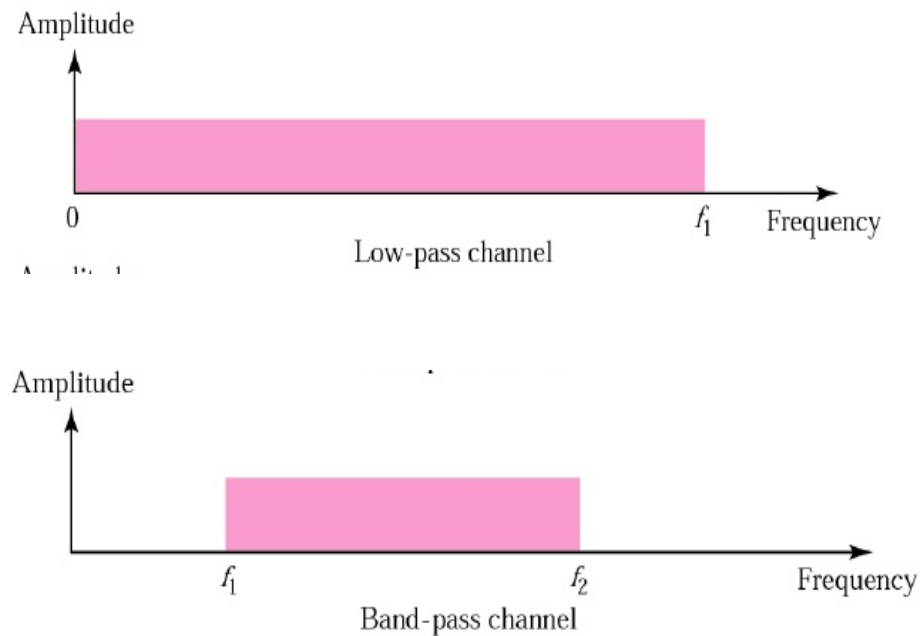
$$\text{SNR} = 10^{(\text{SNR}_{\text{dB}}/10)}$$

$$\text{SNR} = 10^{(36/10)}$$

$$\text{SNR} = 10^{3.6} = 3981$$

Channel Bandwidth

Bandwidth is the difference between the lowest and highest frequency a channel can convey to a receiver.



Channel Capacity

- It is the maximum rate at which data can be correctly communicated over a channel in presence of noise and distortion.
 - is the number of digital values the channel can convey in one second (bps).



Noiseless Channel: Nyquist Bit Rate

For a noiseless channel, the Nyquist bit rate formula defines the **theoretical maximum bit rate of a transmission medium as a function of its bandwidth**

$$C = 2 \times BW \times \log_2 m \quad \text{bits/sec,}$$

C is known as the channel capacity,
BW is the bandwidth of the channel
and m is the number of signal levels used.

Example

- Consider a noiseless channel with a bandwidth of 3kHz transmitting a signal with two signal levels. What is the Nyquist bit rate?

$$C = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

- Consider the same noiseless channel transmitting a signal with four signal levels. What is the Nyquist bit rate?

Noisy Channel: Shannon Capacity

- In reality, we cannot have a noiseless channel; the channel is always noisy.
- In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical **highest data rate for a noisy channel**

$$C = BW \times \log_2 (1 + \text{SNR})$$

Example

- Consider an extremely noisy channel in which the value of the SNR is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0$$

- This means that the capacity of this channel is zero regardless of the bandwidth.
- In other words, we cannot receive any data through this channel.

Example

- A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The SNR is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163 = 3000 \times 11.62 \\ = 34,860 \text{ bps}$$

- This means that the highest bit rate for a telephone line is 34.860 kbps.
- If we want to send data faster than this, we can either increase the bandwidth of the line or improve the SNR

PERFORMANCE

One important issue in networking is the performance of the network. The parameters used to determine the performance of the network are as follows:

1. Bandwidth
2. Throughput
3. Delay (Latency)
4. Jitter

1. Bandwidth

One characteristic that measures network performance is bandwidth. However, the term can be used in two different contexts with two different measuring values: Bandwidth in hertz and bandwidth in bits per second.

Bandwidth in Hertz

Bandwidth in hertz is the range of frequencies contained in a composite signal or the range of frequencies a channel can pass. For example, we can say the bandwidth of a subscriber telephone line is 4 kHz.

Bandwidth in Bits per Seconds

The term bandwidth can also refer to the number of bits per second that a channel, a link, or even a network can transmit. For example, one can say the bandwidth of a Fast Ethernet network (or the links in this network) is a maximum of 100 Mbps. This means that this network can send 100 Mbps.

2. Throughput

- The throughput is a **measure of how fast we can actually send data through a network.**
- Although, at first glance, bandwidth in bits per second and throughput seem the same, they are different.
- **A link may have a bandwidth of B bps, but we can only send T bps through this link with T always less than B .**
- In other words, the bandwidth is a potential measurement of a link; the throughput is an actual measurement of how fast we can send data.
- For example, we may have a link with a bandwidth of 1 Mbps, but the devices connected to the end of the link may handle only 200 kbps.
- This means that we cannot send more than 200 kbps through this link.

Example

➤ A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

➤ Solution:

We can calculate the throughput as

$$\text{Throughput} = \frac{12000 * 10000}{60} = 2 \text{ Mbps}$$

Latency (Delay)

- The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.
- We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.
- **Latency = propagation time + transmission time + queuing time + processing delay**

1. Propagation Delay

Propagation time measures the time required for a bit to travel from the source to the destination. The propagation time is calculated by dividing the distance by the propagation speed.

$$\text{propagation delay (time)} = \frac{\text{Distance}}{\text{Propagation speed}}$$

The propagation speed of electromagnetic signals depends on the medium and on the frequency of the signal. For example, in a vacuum, light is propagated with a speed of 3×10^8 m/s. It is lower in air; it is much lower in cable.

Example

➤ What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4×10^8 m/s in cable.

➤ Solution:

➤
$$\text{Propagation time} = \frac{12000 * 1000}{2.4 * 10^8} = 50 \text{ ms}$$

2. Transmission Time

- In data communications we don't send just 1 bit, we send a message.
- The first bit may take a time equal to the propagation time to reach its destination; the last bit also may take the same amount of time.
- However, there is a time between the first bit leaving the sender and the last bit arriving at the receiver.
- The first bit leaves earlier and arrives earlier; the last bit leaves later and arrives later.
- The time required for transmission of a message depends on the size of the message and the bandwidth of the channel.
- $\text{Transmission Time} = \frac{\text{Message Size}}{\text{Bandwidth}}$

Example

➤ What are the propagation time and the transmission time for a 2.5-kbyte message (an e-mail) if the bandwidth of the network is 1 Gbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.

➤ Solution

➤ *Propagation time* = $\frac{12000 * 1000}{2.4 * 10^8} = 50 \text{ ms}$

➤ *Transmission time* = $\frac{2500 * 8}{10^9} = 0.020 \text{ ms}$

Example

- What are the propagation time and the transmission time for a 5-Mbyte message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s.
- Solution:
- Propagation Time = 50 ms
- Transmission Time = 40s

3. Queuing Time

- The third component in latency is the queuing time, the time needed for each intermediate or end device to hold the message before it can be processed. The queuing time is not a fixed factor; it changes with the load imposed on the network.
- When there is heavy traffic on the network, the queuing time increases.
- An intermediate device, such as a router, queues the arrived messages and processes them one by one. If there are many messages, each message will have to wait.

4. Jitter

- Another performance issue that is related to delay is jitter.
- We can roughly say that jitter is a problem if **different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example).**
- If the delay for the first packet is 20 ms, for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter.